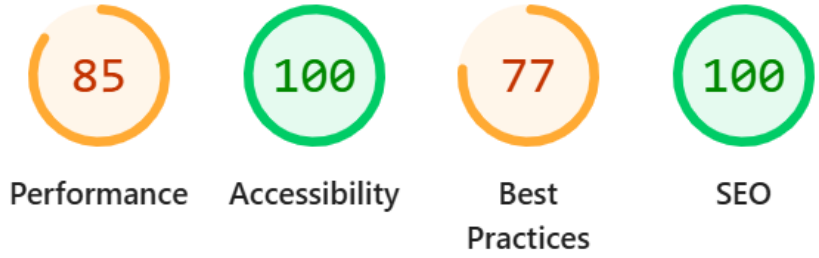
Performance Report

Before Audit

Lighthouse Score



PageSpeed Insights

Performance Issues

1. Large unoptimized images
2. Render-blocking requests
3. Late LCP image discovery
4. Unused CSS
5. No lazy loading or preconnect used for images and iframes

Baseline Metrics

Load time: 1. 4s

Page size: 10.5 MB

LCP (Large Contentful Paint): 5.1s

FCP (First Contentful Paint): 1.1s

Number of Requests: 70

TTI (Time to Interactive): 2,429 ms

Optimizations Implemented

1. Add loading= “lazy” to all 3 album cover images and 2 YouTube iframes

   Before

```html
<div class="album-wrapper">
    <img src="assets/images/FallOff-AlbumCover.png" alt="Two Six Album Cover"
        class="new-music-image">
</div>
```

```html
<div class="video-wrapper">
    <iframe width="560" height="315"
        src="https://www.youtube.com/embed/oVVkmPxGtz8?si=X22Bh4zHg6_6qVqG"
        title="YouTube video player" frameborder="0"
        allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture; web-share"
        referrerpolicy="strict-origin-when-cross-origin" allowfullscreen></iframe>
</div>
```

   After

```html
<div class="album-wrapper">
    <img src="assets/images/FallOff-AlbumCover.png" loading="lazy" alt="Two Six Album Cover"
        class="new-music-image">
</div>
```

```html
<div class="video-wrapper">
    <iframe width="560" height="315"
        src="https://www.youtube.com/embed/oVVkmPxGtz8?si=X22Bh4zHg6_6qVqG"
        title="YouTube video player" frameborder="0"
        allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture; web-share"
        referrerpolicy="strict-origin-when-cross-origin" allowfullscreen loading="lazy"></iframe>
</div>
```

   Why?

   I added lazy loading to all album cover images and iframes, since they were below the fold. By delaying the loading of this content until the user scrolls to it, overall performance will improve.

2. Add preconnect for YouTube

   Before

```html
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-wid
    <title>Forte - Music Journalism</title>

    <!-- SEO META TAGS -->
    <meta name="description"
        content="Forte is a creative digital magazi
    <meta name="keywords" content="music, journalis
    <meta name="author" content="Forte - Music Jour

    <!-- Google Fonts -->
    <link rel="preconnect" href="https://fonts.goog
    <link rel="preconnect" href="https://fonts.gsta
    <link
        href="https://fonts.googleapis.com/css2?fam
        rel="stylesheet">
    <!-- ICONS -->
    <link rel="stylesheet" href="https://cdnjs.clou
        integrity="sha512-2SwdPD6INVrV/lHTZbO2nodKh
        crossorigin="anonymous" referrerpolicy="no-

    <!-- CSS LINK -->
    <link rel="stylesheet" href="style.css">
</head>
```

After

```html
    <!-- YOUTUBE -->
    <link rel="preconnect" href="https://www.youtube.com">
```

Why?

I added preconnect to YouTube for the videos in the interview section so the browser can establish a connection to the server before it's needed. It allows for no connection delay.

3. Add preload for hero image
   Before

```html
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
    <title>Forte - Music Journalism</title>

    <!-- SEO META TAGS -->
    <meta name="description"
        content="Forte is a creative digital magazine
    <meta name="keywords" content="music, journalism,
    <meta name="author" content="Forte - Music Journal

    <!-- Google Fonts -->
    <link rel="preconnect" href="https://fonts.googlea
    <link rel="preconnect" href="https://fonts.gstatic
    <link
        href="https://fonts.googleapis.com/css2?family
        rel="stylesheet">
    <!-- ICONS -->
    <link rel="stylesheet" href="https://cdnjs.cloudfl
        integrity="sha512-2SwdPD6INVrV/lHTZbO2nodKhrnD
        crossorigin="anonymous" referrerpolicy="no-ref

    <!-- YOUTUBE -->
    <link rel="preconnect" href="https://www.youtube.c

    <!-- CSS LINK -->
    <link rel="stylesheet" href="style.css">
</head>
```

After

```html
<!-- HERO IMG -->
<link rel="preload" as="image" href="assets/images/hero-image.jpg">
```

Why?

I added a preload for the hero image, so there can be a higher priority for the image that users will first see. This also allows for shorter connection delay.

4. Unused CSS

Before

```css
/* Font Sizes */
--font-size-hero: clamp(2rem, 5vw + 1rem, 3.5rem);
--font-size-section-lg: clamp(1.75rem, 4vw + 0.5rem, 2.5rem);
--font-size-section: clamp(1.5rem, 3vw + 0.5rem, 2rem);
--font-size-card-title: clamp(1.1rem, 2vw + 0.5rem, 1.35rem);
--font-size-body: clamp(0.9rem, 1vw + 0.5rem, 1rem);
--font-size-small: clamp(0.85rem, 1vw + 0.5rem, 0.95rem);
--font-size-label: 0.7rem;
--font-size-meta: 0.5rem;
```

```
/* Shadows */
--shadow-card: 0 4px 20px rgba(0, 0, 0, 0.08);
--shadow-card-sm: 0 2px 12px rgba(0, 0, 0, 0.06);
--shadow-hover: 0 4px 16px rgba(0, 0, 0, 0.01);
```

```
/* Layout Max Widths */
--max-width-narrow: 700px;
--max-width-content: 800px;
--max-width-wide: 1000px;
```

After

```
/* Font Sizes */
--font-size-hero: clamp(2rem, 5vw + 1rem, 3.5rem);
--font-size-section-lg: clamp(1.75rem, 4vw + 0.5rem, 2.5rem);
--font-size-section: clamp(1.5rem, 3vw + 0.5rem, 2rem);
--font-size-card-title: clamp(1.1rem, 2vw + 0.5rem, 1.35rem);
--font-size-body: clamp(0.9rem, 1vw + 0.5rem, 1rem);
--font-size-small: clamp(0.85rem, 1vw + 0.5rem, 0.95rem);
--font-size-label: 0.7rem;
```

```
/* Shadows */
--shadow-card: 0 4px 20px rgba(0, 0, 0, 0.08);
--shadow-card-sm: 0 2px 12px rgba(0, 0, 0, 0.06);
```

Why?

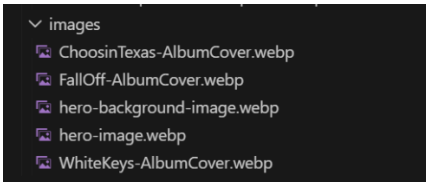
I removed font-size-meta, shadow-hover, and all layout max-widths from the root, as they were not used in any CSS. By removing unused CSS, the browser does not have to spend time downloading unnecessary information and can instead focus on the CSS in use.

5. Compress Images to WebP

Before

```
images
  ChoosinTexas-AlbumCover.png
  FallOff-AlbumCover.png
  hero-background-image.jpg
  hero-image.jpg
  WhiteKeys-AlbumCover.png
```

After

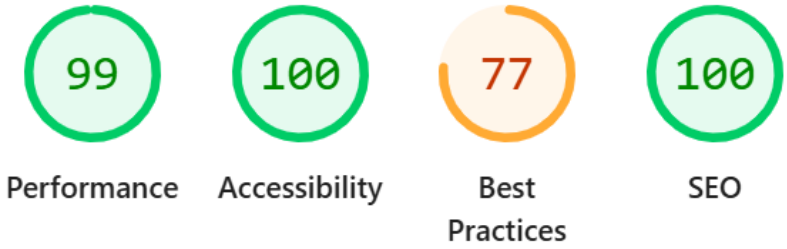


Why?

I converted all the images used to WebP to compress the file size. Reducing the page size allows the page to be downloaded faster, immediately improving performance.

After Audit

Lighthouse Score



PageSpeed Insights

Final Metrics

Load time: 275ms

Page size: 4.7 MB

LCP (Large Contentful Paint): 0.8s

FCP (First Contentful Paint): 0.6s

Number of Requests: 20

TTI (Time to Interactive): 777ms

Comparison Table

	Before	After
Load Time	1.4s	275ms
Page Size	10.5MB	4.7MB
LCP	5.1s	0.8s
FCP	1.1s	0.6s
Number of Requests	70	20
TTI	2, 429ms	777ms

Reflection

The biggest impact was changing the images from PNG to WebP, which drastically reduced page size, allowing the browser to load the page faster. Even though I had already reduced the image size twice using Squoosh and ImageOptim for the accessibility audit, converting them to WebP made the greatest difference. I was shocked by how leaving unused CSS in your sheet can reduce performance and how much unused CSS was in my code. For future projects, I have learned how important image, audio, and video file sizes are to performance. I will be more diligent about ensuring that I am compressing the amount of storage I use.